

Scott Anthony Barlowe

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RESEARCH INTERESTS

Computer Science Education, Visual Analytics, Applications of Machine Learning/AI

EDUCATION

Doctor of Philosophy, Computing and Information Systems

December 2011, University of North Carolina – Charlotte (Charlotte, NC)

Dissertation: *A Visual Analytics Approach to Feature Discovery and Subspace Exploration in Protein Flexibility Matrices*

Master of Science, Computer Science

August 2005, Appalachian State University (Boone, NC)

Thesis: *A Lifting-Based Knowledge Discovery in Microarray Data*

Bachelor of Science, Industrial Technology, summa cum laude

December 1996, Appalachian State University (Boone, NC)

Concentration: Electronics Technology Minor: Business

PROFESSIONAL EXPERIENCE

Professor, Department of Mathematics and Computer Science

Western Carolina University Cullowhee, NC

July 2024 – Present

Associate Professor, Department of Mathematics and Computer Science

Western Carolina University Cullowhee, NC

July 2019 – July 2024

Assistant Professor, Department of Mathematics and Computer Science

Western Carolina University Cullowhee, NC

August 2013 – July 2019

Program Head and Instructor, Simulation and Game Development

Stanly Community College Albemarle, NC

April 2006 – July 2013

Instructor of Record / Graduate Teaching Assistant, Department of Computer Science

Appalachian State University Boone, NC

May 2002 – May 2005

ACADEMIC SERVICE

Western Carolina University (August 2013 – Present)

Leadership

Program Director, Computer Science (July 2016 – June 2019, August 2023 - Present)

Chair, Search Committee for tenure-track position in Computer Science (May 2017 – July 2017, Oct 2024 – March 2025)

Instruction

CS 495/496, Capstone

Description: Responsible for advising two-semester projects as independent studies. Groups (1 – 2 students per group) develop proposals, biweekly progress reports, final reports, and final presentations each semester. Research results in a required poster at a departmental poster session.

Projects Supervised: (* indicates work towards peer-reviewed scholarly works, † indicates abstracts that were selected for oral presentation after a double-blind review by an interdisciplinary university committee.)

- * *Integrating Machine Learning for Sign Language Recognition in Video* (Z. Eanes and A. Charlot, 2025)
- *A Visualization System for Exploring Warehouse Inventory* (T. Freeman, M. Tucker, and J. Mullins, 2025)
- *An Image-based App for Learning Sign Language* (M. Agudelo Ganan, M. Talle, and T. Holmberg, 2025)
- *Immersive Visualization of Game Statistics with the Oculus Quest* (R. Burkhalter and S. Kelly, 2024)
- * *Mining Cryptocurrency Transaction Graphs* (K. Patel and N. Hassett, 2024)
- *Exploring Game Data from Steam* (C. Farrell, 2023)
- *A Pipeline for Weather Data Exploration* (C. Fowler and Z. Laney, 2023)
- * *A Visual Analytics Approach to Exploring Bitcoin Transactions* (A. Kirk and K. Anderson, 2023)
- *Node-View: Visualizing Bitcoin with Multiple Views* (J. Strong and N. Wofford, 2023)
- *Data Visualization with the Hololens* (A. Shelton and Tyler Rogers, 2023)
- * *Visualizing and Storing Software Defined Networks, Part II* (M. Stanley and N. Lowry, 2022)
- *Visualizing the Internet Movie Database* (D. Gorlesky and B. Dale, 2021)
- * *Visualizing and Storing Software Defined Networks, Part I* (G. Cruz and C. Tipton, 2021)
- * *Vocabulo: Mobile AI for Language Acquisition* (J. Ginn and C. Bautista Salas, 2020)
- *Visualizing North Carolina Cancer Statistics* (N. Ruby and M. Massey, 2019)
- *A Comprehensive Toolkit for Steganography and Image Processing* (S. Towne and J. Caplinger, 2019)
- *A Guided Tour of Cullowhee with Image Processing* (J. Landolf, 2018)
- *Visualization of Aggregated Death Data* (A. Williams, 2018)
- *Visual Analysis of Regional Tourism Data in Western North Carolina* (J. Brink, 2018)
- *Asteroids: A 3D Kinect-Controlled Video Game* (K. Pflug and J. Mechanye, 2018)
- *A Gesture-based 3D Arcade Game* (J. Sims and E. Arroyo, 2017)
- † *SketchCAD-A Web-based 3D Modeling Tool* (C. Hreha and C. Pero, 2016)
- *Teaching Encryption with the Lock and Key Encryption Tool* (T. White, 2016)
- † *Towards a Multipurpose Suite of Encryption Tools* (C. Bryson, 2016)
- *ShadeIT: An Interactive Platform for Developing Shading Programs* (M. Anger, 2016)
- *Visual Analysis of Fluid Flow Simulations* (T. Hardin, 2015)
- *Visualizing StarCraft 2 Match and Ladder Statistics* (C. Griser, 2015)
- *WaveDraw: A Gesture-based Drawing Program* (B. Tetrault, 2015)
- *A Visual Tool for Understanding Graph Algorithms* (D. Dameron, 2015)
- *Visual Analysis of Web Accessible Game Statistics* (J. Stilwell and S. Facello, 2015)

CS 475, Cloud Computing, 3 credit hours

CS 465, Computer Networking, 3 credit hours

CS 461, Computer Graphics, 3 credit hours

CS 353, Professional Ethics in Computing, 1-2 credit hours

CS 351, Data Structures, 3-4 credit hours

CS 260, Computer Organization and Design

CS 210, Internet Security and Ethics, 3 credit hours

CS 192, First Year Seminar - Applied AI, Programming and Ethics, 3 credit hours

CS 191, First Year Seminar – How Does Software Work?, 3 credit hours

CS 151, Problem Solving and Programming II, 4 credit hours

CS 150, Problem Solving and Programming I, 4 credit hours

University Service

Division of IT AI Governance Committee (Spring 2026 – Present)

AI Working Group (Fall 2023 – May 2025)

Board of Governors Excellence in Teaching Award Committee (Fall 2022 – Fall 2024)

LMS Governance Committee (Fall 2019 – Present)

University Infrastructure Advisory Committee (Fall 2016 - Spring 2017)

MOOC (Massive Open Online Course) Feasibility Committee (Fall 2013 – Spring 2014)

College Service

College of Engineering and Technology Collegial Review Committee (External Member, Fall 2025)

Apodaca Science Building Steering Committee (Spring 2016 – Summer 2021)

Department Service

Elected Positions

Tenure and Promotion Review Committee (Fall 2019 – Spring 2025)

Post-Tenure Review Committee (Fall 2022 – Spring 2023)

Annual Faculty Evaluation Committee (2020, 2021, 2022, 2025)

Other

Computer Science Committee (Fall 2013 – present)

CS Lab (Apodaca) Administrator (Fall 2021 – present)

Search Committee for Administrative Support Associate (Summer 2023)

Departmental Collegial Review Document Revision Committee, ad-hoc (Spring 2023)

Search Committee for tenure-track position in Computer Science (November 2022 – April 2023)

Departmental Collegial Review Document Revision Committee, ad-hoc (Spring 2016)

Co-Advisor, ACM/Computer Science Club (Fall 2013 – Spring 2015)

Stanly Community College (April 2006 – August 2013)

Leadership

Program Director, Simulation and Game Development (April 2006 – August 2013)

Instruction

SGD 285, Software Engineering

SGD 214, Advanced 3D Modeling
SGD 213, Simulation and Game Programming II
SGD 212, Game Design II
SGD 125, Artificial Intelligence
SGD 114, 3D Modeling
SGD 112, Game Design I
SGD 113, Simulation and Game Programming I
SGD 111, Introduction to Simulation and Game Development
CTS 289, System Support Project
CTS 285, Systems Analysis and Design
CIS 115, Programming and Logic
CIS 110, Introduction to Computers

Committee Membership

Faculty senate representative, elected to 2-year term (2010 – 2012) and served as Secretary both years.
Quality Enhancement Plan Committee (Summer 2012 – Winter 2012).

Appalachian State University (May 2002 – May 2005)

Instruction

CS 1410, Introduction to Computers
CS 2440, Computer Science II Laboratory

REFEREED SCHOLARLY WORKS (* indicates leadership in cross-disciplinary projects, † indicates undergraduate students under my mentorship)

[24] †Z. Eanes, **S. Barlowe**, †A. Charlot, and A. Scott. *A Novel Hybrid Framework for Mobile Sign Language Translation, Local LLM Text Generation, and Analytics*. Journal of Computing Sciences in Colleges, November 2025.

[23] **S. Barlowe**, †D. Aoulou, and †A. Ponce-Castillo. *WIP: Generative AI as an Instructional Resource in a Computer Science Ethics Course*. IEEE FIE 2024.

[22] **S. Barlowe**, †K. Patel, †N. Hassett, †K. Anderson, and †A. Kirk. *Enhancing Visual Exploration of Illicit Bitcoin Transactions with Machine Learning*. IEEE AIXB 2024.

[21] **S. Barlowe** and A. Scott. *News Consumption Among CS Majors: Habits, Perceptions, and Challenges*. Proceedings of the 2023 ACM Southeast Conference. p. 162–166, April 2023.

[20] **S. Barlowe**, †M. Stanley, †N. Lowry, †C. Tipton, and †G. Cruz. *Towards An Integrated Framework for Managing Software-Defined Networking Models*. SoutheastCon 2023, Orlando, FL, USA, 2023, p. 23-30.

[19] **S. Barlowe**. *Encouraging Global Awareness in a Computer Networking Course*. IEEE Transactions on Education, (64) 2, p. 195-201, May 2021.

- [18] †J. Ginn, †C. Bautista Salas, ***S. Barlowe**, and W. Lehman. *A Novel Framework for Integrating Mobile Machine Learning in L2 Vocabulary Acquisition*. Journal of Computer Sciences in Colleges, (36) 5, p. 47-56. Jan. 2021.
- [17] A. Scott, W. Krehling, M. Holliday, and **S. Barlowe**. *A Holistic Capstone Experience: Beyond Technical Ability*. 18th Annual Conference on Information Technology Education 2017, p. 1-6.
- [16] ***S. Barlowe**, H. Coan, and R. Youker. *SubVis: An Interactive R Package for Exploring the Effects of Multiple Substitution Matrices on Pairwise Sequence Alignment*. PeerJ June 27, 2017, doi: 10.7717/peerj.3492.
- [15] A. Scott and **S. Barlowe**. *How Software Works - Computational Thinking and Ethics Before CS1*. IEEE Frontiers in Education 2016, p. 1-9.
- [14] C. Zhang, J. Yang, F.B. Zhan, X. Gong, J. Brender, P. Langlois, **S. Barlowe**, and Y. Zhao. *A Visual Analytics Approach to High-Dimensional Logistic Regression Modeling and its Application to an Environmental Health Study*. Proc. IEEE Pacific Visualization Symposium, 2016, p. 136-143. (31% acceptance rate)
- [13] **S. Barlowe** and A. Scott. *O-Charts: Towards an Effective Toolkit for Teaching Time Complexity*. IEEE Frontiers in Education 2015, p. 1-4.
- [12] C. Xie, W. Chen, X. Huang, Y. Hu, **S. Barlowe**, J. Yang. *VAET: A Visual Analytics Approach for E-transactions Time-series*. IEEE Visual Analytics Science and Technology, 2014, (12):1743-52. (37% acceptance rate)
- [11] Y. Hu, X. Huang, C. Zhang, Y. Wu, J. Yang, Y. Zhao, **S. Barlowe**, W. Chen. *CateDocs: A New Visual Analytics Approach to Exploring High Dimensional Categorical Datasets*. IEEE VisWeek 2013. (2-page paper in the electronic proceedings, presented as poster. Abstract located at <https://ieevis.org/year/2013/poster/catedocs-new-visual-analytics-approach-exploring-high-dimensional-categorical-datas>).
- [10] C. Zhang, J. Yang, F.B. Zhan, X. Gong, J.D. Brender, P. Langlois, **S. Barlowe**. *Identifying Risk Factors for Birth Defects in High Dimensional Environmental Health Data*. IEEE VisWeek 2013. (2-page paper in the electronic proceedings, presented as poster. Abstract located at <https://ieevis.org/year/2013/poster/identifying-risk-factors-birth-defects-high-dimensional-environmental-health-data>).
- [9] ***S. Barlowe**, J. Yang, D. J. Jacobs, D. R. Livesay, J. Alsakran, Y. Zhao, D. Verma, J. Mottonen. *A Visual Analytics Approach to Exploring Protein Flexibility Subspaces*. Proc. IEEE Pacific Visualization Symposium, 2013, p. 193-200. (29% acceptance rate)
- [8] J. Yang, Y. Liu, X. Zhang, X. Yuan, Y. Zhao, **S. Barlowe**, S. Liu. *PIWI: Visually Exploring Graphs Based on Their Community Structure*. Transactions on Visualization and Computer Graphics, 19(6), p. 1034-1047 Jun 2013..
- [7] Y. Liu, **S. Barlowe**, Y. Feng, J. Yang, M. Jiang. *Evaluating exploratory visualization systems: A user study on how clustering-based visualization systems support information seeking from large document collections*. Information Visualization, 12(1), p. 25 – 43, October 2012.
- [6] Y. Chen, J. Alsakran, **S. Barlowe**, J. Yang, Y. Zhao. *Supporting Effective Common Ground Construction in Asynchronous Collaborative Visual Analytics*. IEEE Visual Analytics Science and Technology, 2011. (32% acceptance rate)
- [5] ***S. Barlowe**, Y. Liu, J. Yang, D. R. Livesay, D. J. Jacobs, J. Mottonen, and D. Verma. *WaveMap: Interactively Discovering Features From Protein Flexibility Matrices Using Wavelet-based Visual Analytics*. Computer Graphics Forum, (30) p. 1001-1010, 2011. (28% acceptance rate)

- [4] Y. Chen, **S. Barlowe**, J. Yang. *Click2Annotate: Automated Insight Externalization with Rich Semantics*. IEEE Visual Analytics Science and Technology, pp. 155-162, 2010. (29% acceptance rate)
- [3] Y. Chen, J. Yang, **S. Barlowe**, and D.H. Jeong. *Touch2Annotate - Generating better annotations with less human efforts on multi-touch interfaces*. CHI Work-In-Progress, p. 3703–3708, 2010.
- [2] **S. Barlowe**, T. Zhang, Y. Liu, J. Yang and D. J. Jacobs. *Multivariate Visual Explanation for High Dimensional Datasets*. IEEE Symposium on Visual Analytics Science and Technology, p. 147 – 154, 2008. (34% acceptance rate)
- [1] **S. Barlowe** and R. Tashakkori. *A Lifting-Based Knowledge Discovery in Microarray Data*. Proceedings of the 2005 International Conference on Mathematics and Engineering Techniques in Medicine and Biological Sciences, p. 291-298, 2005.

BOOK CHAPTERS

J. Yang and **S. Barlowe**. *A Dimension Management Framework for High Dimensional Visualization*. Advances in Information and Intelligent Systems, Vol. 251, 2009.

GRANT ACTIVITY

National Science Foundation

Submitted by R. Tashakkori, Department of Computer Science at Appalachian State University (lead institution)

AppEd platform to support game-based learning and research

Principle Investigator/Director at Western Carolina University

Fall 2022, **\$202,994** (WCU's part) [Not Funded]

North Carolina Board of Science, Technology, and Innovation / Department of Commerce

What is the distribution of data literacy secondary school pathways and higher education certificate and degree programs by NC county and tier?

Principle Investigator

Fall 2022 – Spring 2023, **\$6720** [Funded]

STEM-Entrepreneurship Classroom Grants

The Cherokee Preservation Foundation (WNCedNet)

Program Co-Director

July 2014, **\$790** [Funded]

NC Community College System

Program Director

December 2006 – June 2007, **\$124,859** [Funded]

REVIEWING

Consortium on Computing Sciences in Colleges	2021, 2022, 2023
IEEE VIS	2021, 2022
IEEE Transactions on Education	2020
Computer Animation and Virtual Worlds	2019
Eurographics/ IEEE-VGTC Symposium on Visualization	2012, 2013, 2014, 2015, 2016
IEEE PacificVis	2013, 2015, 2016
Visualization and Data Analysis	2014
Information Visualization	2013, 2014
BioVis	2013, 2014

INVITED PRESENTATIONS

North Carolina Office of Strategic Partnerships

Panel Member

Monthly Connect (virtual, 150+ attendees)

February 23, 2023

Fourth Quarter Meeting of The NC Board of Science, Technology & Innovation

Presenter

November 11, 2022

CONFERENCE PRESENTATIONS

S. Barlowe. *A Novel Hybrid Framework for Mobile Sign Language Translation, Local LLM Text Generation, and Analytics.* Journal of Computing Sciences in Colleges, November 2025.

S. Barlowe, *WIP: Generative AI as an Instructional Resource in a Computer Science Ethics Course.* IEEE FIE 2024.

S. Barlowe, *Enhancing Visual Exploration of Illicit Bitcoin Transactions with Machine Learning.* IEEE AIXB 2024.

S. Barlowe. *News Consumption Among CS Majors: Habits, Perceptions, and Challenges.* ACM Southeast Conference. April 12, 2023. Virtual.

S. Barlowe. *Towards An Integrated Framework for Managing Software-Defined Networking Models* IEEE Southeastcon 2023. April 1, 2023. Virtual.

S. Barlowe. *A Novel Framework for Integrating Mobile Machine Learning in L2 Vocabulary Acquisition.* Consortium for Computing Sciences in Colleges 2023 (CCSC: 2023). Jan. 22, 2021. Virtual.

S. Barlowe. *O-Charts: Towards an Effective Toolkit for Teaching Time Complexity.* IEEE Frontiers in Education 2015, October 24, 2015. El Paso, Texas.

S. Barlowe. *Multivariate Visual Explanation for High Dimensional Datasets.* IEEE Symposium on Visual Analytics Science and Technology, Oct. 22, 2008. Columbus, Ohio.

WORKSHOP PRESENTATIONS

Scratch Programming

Student Technology Advisory Council (STAC) Spring Conference
Cherokee Central Schools, April 22, 2015

Getting Fruity with Game Maker Objects

Student Technology Advisory Council (STAC) Fall Conference
Cherokee Central Schools, September 18, 2014
40 students over 2 sessions, 90 minutes each

Video Game Building Mini-Camp

(co-director with Jackie Methven of Jackson County Schools)
Smokey Mountain High School, Sylva, NC, July 2014
5-day workshop with 13 high school students
Funded by STEM-Entrepreneurship Classroom Grants
and The Cherokee Preservation Foundation (WNCedNet)

Scott A. Barlowe

An Introduction to Jython

Carolina Coding Initiative, December 2013
North Carolina Center for the Advancement of Teaching
3 workshops (1 with teachers, 2 with general public)

**It's Game On! With Level Design and
Building 2d Arcade Games with Game Maker**

Free College Day Workshop on Game Development
Stanly Community College, June 23, 2012

Game Maker: An Alternative Platform For Teaching CIS 115

NC Computer Instructor's Association.
Stanly Community College, Feb 2008

PROFESSIONAL DEVELOPMENT

Course, *Probabilistic Graphical Models 1: Representation*, Stanford University / Coursera. Certificate.
2021.

Google Cloud Platform Badges, Google Cloud Platform. 2021.

Badges: Networking in Google Cloud
Baseline: Data, ML, AI
Baseline: Infrastructure
Google Cloud Fundamentals

Course: *Bioinformatics: Life Sciences on Your Computer*, John Hopkins University / Coursera. Certificate.
2014.

AWARDS / NOMINATIONS

Finalist, Innovative Scholarship Award (Western Carolina University, Sp 2021)

**Named by Chancellor's List recipient as the faculty member who most supported their academic
success the previous semester** (Western Carolina University, Sp 2015)

First Year Experience Advocate Nominee (Western Carolina University, Sp 2014/Sp 2016)

Outstanding Faculty Member Award (Stanly Community College, 2009, \$250)